

Supporting Information

Enhanced dielectric breakdown property of polypropylene based on mesoscopic structure modulation by crystal phase transformation for high voltage power cable insulation

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2.2. Sample Preparation

Composition of the samples

Table S1. Composition of the samples

Sample	Component (wt.%)	
	iPP (T30)	β -NA (WBG-II)
PP-p	100	0
PP- β -0.05	99.95	0.05
PP- β -0.1	99.9	0.1
PP- β -0.2	99.8	0.2
PP- β -0.4	99.6	0.4

2.4. Mesoscopic Electrical Tree Degradation

Mesoscopic electrical tree test platform

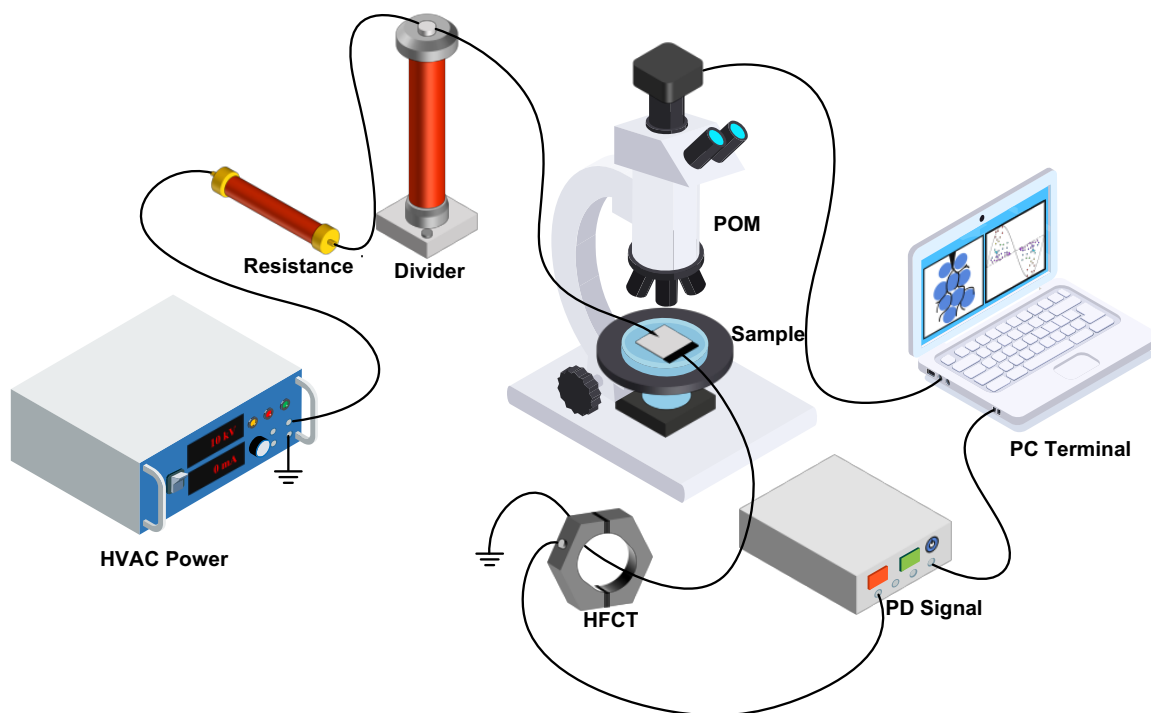


Figure S1. mesoscopic electrical tree test platform

2.6. Molecular Dynamics Simulation

Equations of relative permittivity

$$\Phi(t) = \langle M(0) \cdot M(t) \rangle \quad (S1)$$

where $M(t)$ is the dipole moment of the molecular chain at the moment t , and its value is the dot product of charge quantity and its position vector of the molecular chain at that moment. $\langle M(0) \cdot M(t) \rangle$ is the average of the sum of the number products of $M(0)$ and $M(t)$ for all molecular chains in the system^{1,2}.

$$\frac{\varepsilon(\omega) - \varepsilon_\infty}{\varepsilon_r - \varepsilon_\infty} = \int - \left(\frac{d\Phi(t)}{dt} \right) \cos(\omega t) dt \quad (S2)$$

where ε_r is the static permittivity, representing the value of the permittivity of the system when the frequency tends to 0 at infinity. $\varepsilon(\omega)$ is the value of the permittivity as the frequency varies. ε_∞ is the optical frequency permittivity, representing the value of the permittivity when the frequency is infinity. For polypropylene, the value of the optical frequency permittivity is about 1.9^{3,4}.

$$\Delta\varepsilon = \frac{4\pi}{3} \frac{\langle M(0)^2 \rangle - \langle M(0) \rangle^2}{V \varepsilon_0 k_B T} \quad (S3)$$

where $\Delta\varepsilon$ is the relaxation strength, $\Delta\varepsilon = \varepsilon_r - \varepsilon_\infty$, V is the volume of the system, ε_0 is the vacuum permittivity, k_B is the Boltzmann constant, T is the temperature of the system, $\langle M(0)^2 \rangle$ is the mean-square dipole moment of the dipole in the system, and $\langle M(0) \rangle^2$ is the squared value of the average dipole moment of the dipole in the system⁵.

2.7. Finite Element Analysis

Voronoi diagram distribution principle

Voronoi diagram is a geometric concept and represents a spatial segmentation algorithm. Its distribution principle is often used as the modeling basis of crystal material structure, and it has the characteristics of discreteness and locality⁶.

Among them, in the application of the mesoscopic structure model of polypropylene, the discreteness ensures that each spherulite exists in an independent form at the initial stage of crystallization. Meanwhile, the locality is reflected in the later stage of crystallization, which is

restricted by the limited spatial range. The size of any spherulite will be affected by the neighboring spherulites during the growth process.

Specifically, in the modeling process of this paper, according to the experimental results in Figure 2, the space ratio of crystalline region to amorphous region can be set in equal proportion, which is also applied and reflected in the modeling process in Figure 7.

3. Results

Equations of crystallinity

$$X_{m\alpha} = \frac{\Delta H_{m\alpha}}{\Delta H_{m\alpha}^0} \quad (S4)$$

$$X_{m\beta} = \frac{\Delta H_{m\beta}}{\Delta H_{m\beta}^0} \quad (S5)$$

$$X_m = X_{m\alpha} + X_{m\beta} \quad (S6)$$

where $\Delta H_{m\alpha}$ is the melting enthalpy of the α -crystal melting peak, $\Delta H_{m\beta}$ is the melting enthalpy of the β -crystal melting peak, $\Delta H_{m\alpha}^0$ is the melting enthalpy of the α -crystal when it is fully crystallized, and its value is 177.0 J/g. $\Delta H_{m\beta}^0$ is the melting enthalpy of the β -crystal when it is fully crystallized, and its value is 168.5 J/g^{7,8}.

Crystallinity and relative crystallization percentage

Table S2. Crystallinity X_m and relative crystallization percentage K of various samples

Sample	X_m (%)	K (%)	
		α -crystal	β -crystal
PP-p	40.52	100	0
PP- β -0.05	44.11	64.01	35.99
PP- β -0.1	46.89	40.60	59.40
PP- β -0.2	49.76	22.28	77.72
PP- β -0.4	47.44	10.79	89.021

Avrami equation

$$\log[-\ln(1 - X(t))] = n \log t + \log Z_t \quad (S7)$$

Where t is the isothermal crystallization time, $X(t)$ is the change of relative crystallinity of the sample with time in the isothermal crystallization process, Z_t is the crystallization rate, n is the Avrami exponent, the magnitude of the value is related to the mechanism of nucleation and the way of growth. The smaller the value, the more pronounced the heterogeneous nucleation effect and the greater the number of nuclei. $t_{1/2}$ is the half-crystallization time, which corresponds to the time required when $X(t) = 50\%$ ⁹.

PRPD spectrums

Figure S2 shows the partial discharge signal inside the sample under the action of AC voltage. Where Figure S2(a) shows the raw data. For further distinguishing the differences of partial discharge in different quadrants, the polar coordinate system as in Fig. S2(b) is used in this study for representation. Among them, the length of the radius of the circle characterizes the magnitude of the partial discharges, and the color gradually becomes darker from the inner to the outer, which corresponds to a gradual increase in the magnitude of the discharges. The red ray corresponds to the partial discharge occurring at this phase angle, and its length is the maximum value of the discharge. The yellow shading represents the range of the sinusoidal voltage signal in polar coordinates.

The traditional drawing form of phase-resolved partial discharge spectra is similar to Figure S2(a), in which the discharge information exists in the form of scattered points in the two-dimensional coordinate system. However, there are two obvious disadvantages in this drawing method. On the one hand, it is impossible to distinguish the difference of partial discharge intensity at specific phase points. On the other hand, there is a lack of intuitive contrast between different quadrants.

For solving the above-mentioned problems, a new phase-resolved partial discharge spectra drawing method is proposed in this study, as shown in Figure S2(b). In this drawing form, we can not only visually compare the single maximum discharge in different phases, but also effectively observe the fluctuation of partial discharge under the action of sinusoidal voltage signals in different quadrants.

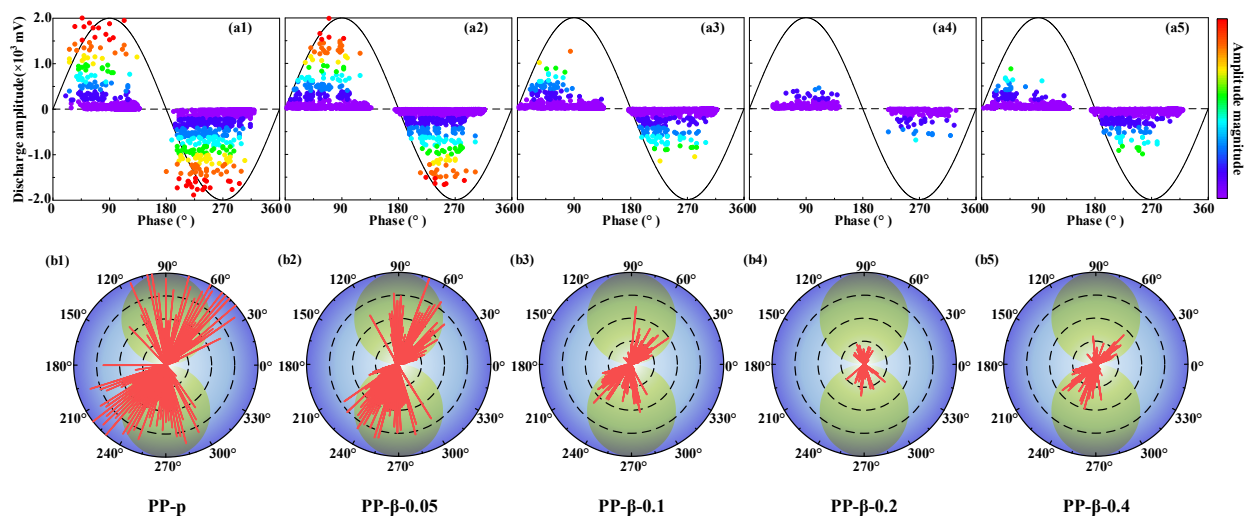


Figure S2. Partial discharge (a) raw data, and (b) phase-resolved partial discharge spectrums obtained after processing.

Binarized electrical tree images before and after processing.

In order to quantify the degradation degree of the electrical tree within the sample, while avoiding the interference of the crystalline morphology of the material on the accuracy of the statistical results, the morphological image of the electrical tree in Figure S3(a) was binarized. The principle of this image processing method is to exclude interfering distortion filtering by specifying a grayscale threshold, thus obtaining a binarized image showing only the electric tree channel as shown in Figure S3(b)¹⁰.

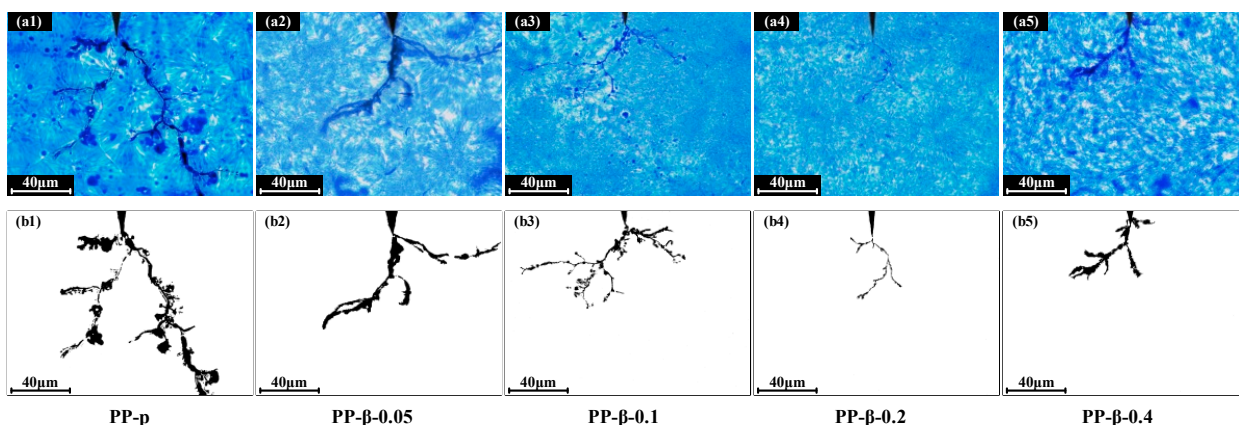


Figure S3. (a) mesoscopic electrical tree images, (c) binarized electrical tree images obtained after processing.

Equation of Weibull distribution

$$P(E) = 1 - \exp \left[- \left(\frac{E}{\alpha} \right)^\beta \right] \quad (\text{S8})$$

Where α is a scale parameter, corresponding to the value of AC breakdown strength when the breakdown probability is 63.2%, β is the shape parameter, which has an inverse relationship with the dispersion of the data, $P(E)$ is the value of the breakdown probability; and E is the value of the breakdown strength when the sample breaks down¹¹.

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